

SSM-WM Architecture for Humanoid Robot State Prediction and MPC

$O(T \log T)$

$O(1)$ step

0.24M params

Code-aligned method overview: state-action encoding, gated diagonal SSM memory, prediction rollout, and MPC planning

data flow

control loop

1 Input history

Humanoid robot trajectories

states
 $s_{\{t-T:t\}}$

actions
 $a_{\{t-T:t-1\}}$
concat $[s_t; a_t]$

2 State-action encoder

Project each time step to latent tokens

Linear $d_s+d_a \rightarrow D$

GELU activation

Linear $D \rightarrow D$

latent sequence $z_{\{0:T\}}$

3 Gated Diag-SSM backbone

Stack L blocks for long-range temporal memory

SSM Block 1

LN

DiagSSM

Gate

hidden memory

4 Prediction head

Decode last hidden token

LayerNorm

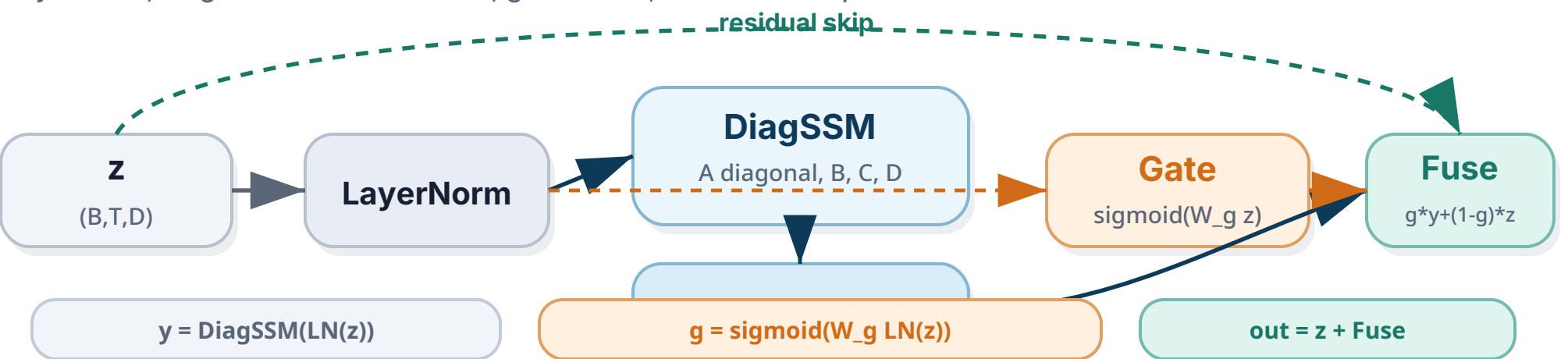
FC + GELU + FC

residual state add

$\hat{s}_{t+1} = s_t + \Delta s$

Gated Diag-SSM block detail

LayerNorm, diagonal SSM convolution, gate fusion, and residual update.



Rollout and MPC usage

Predictions support training loss and online planning.

Training

single-step MSE
+ H-step rollout

MPC

optimize actions
execute first action

receding horizon

Key message: lightweight SSM memory replaces expensive sequence modeling, while the predicted next state closes the MPC control loop.